

CSA0251

C Programming for recursion

HOMEWORK - 3

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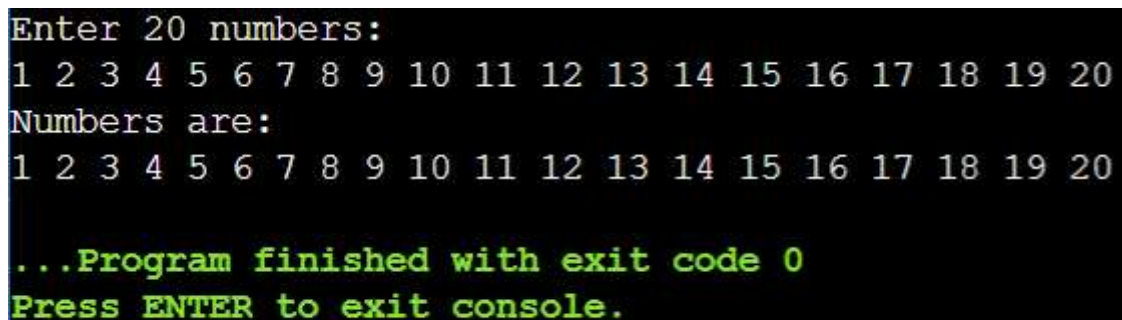
1. Write a C program to Read and display 20 numbers using array and run time input.

PROGRAM :

```
#include <stdio.h>

int main() {
    int a[20], i;
    printf("Enter 20 numbers:\n");
    for(i=0;i<20;i++)
        scanf("%d",&a[i]);
    printf("Numbers are:\n");
    for(i=0;i<20;i++)
        printf("%d ",a[i]);
    return 0;
}
```

OUTPUT:

A screenshot of a terminal window showing the output of the C program. The text is as follows:
Enter 20 numbers:
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
Numbers are:
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
...Program finished with exit code 0
Press ENTER to exit console.
The first two lines of output are in a light blue font, while the last three lines are in a light green font.

```
Enter 20 numbers:
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
Numbers are:
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
...Program finished with exit code 0
Press ENTER to exit console.
```

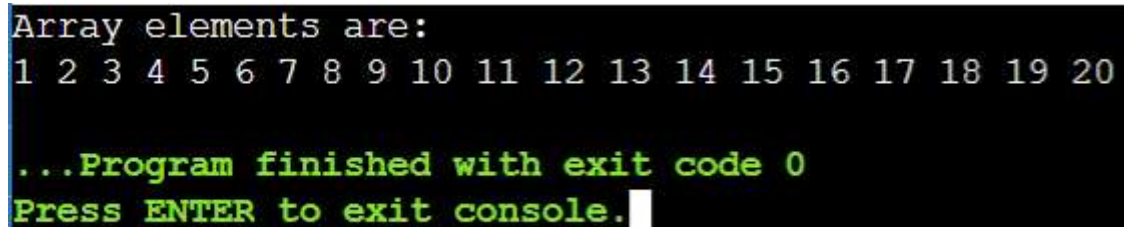
2. Write a C program to Get 20 numbers using array initialization and display these 20 numbers using array.

PROGRAM:

```
#include <stdio.h>

int main() {
    int a[20] = {
        1,2,3,4,5,
        6,7,8,9,10,
        11,12,13,14,15,
        16,17,18,19,20
    };
    int i;
    printf("Array elements are:\n");
    for(i = 0; i < 20; i++) {
        printf("%d ", a[i]);
    }
    return 0;
}
```

OUTPUT :



```
Array elements are:
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
...Program finished with exit code 0
Press ENTER to exit console.
```

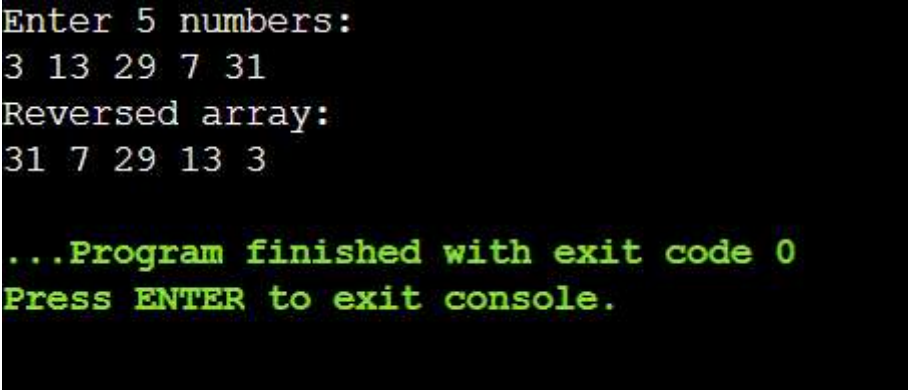
3. Write a program to reverse the elements of an array.

PROGRAM:

```
#include <stdio.h>

int main() {
    int a[5], i;
    printf("Enter 5 numbers:\n");
    for(i=0;i<5;i++)
        scanf("%d",&a[i]);
    printf("Reversed array:\n");
    for(i=4;i>=0;i--)
        printf("%d ",a[i]);
    return 0;
}
```

OUTPUT:



```
Enter 5 numbers:
3 13 29 7 31
Reversed array:
31 7 29 13 3

...Program finished with exit code 0
Press ENTER to exit console.
```

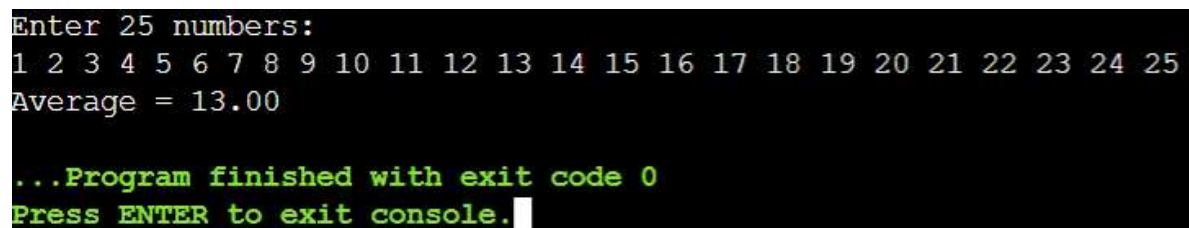
4. Write a C program to find the average of 25 numbers using array.

PROGRAM:

```
#include <stdio.h>

int main() {
    int a[25], i, sum=0;
    printf("Enter 25 numbers:\n");
    for(i=0;i<25;i++) {
        scanf("%d",&a[i]);
        sum += a[i];
    }
    printf("Average = %.2f", (float)sum/25);
    return 0;
}
```

OUTPUT:

A screenshot of a terminal window showing the execution of the C program. The first line is the prompt 'Enter 25 numbers:'. The second line shows 25 numbers from 1 to 25, each followed by a space. The third line shows the calculated average 'Average = 13.00'. The fourth line shows the program completion message '...Program finished with exit code 0'. The fifth line shows the prompt 'Press ENTER to exit console.' followed by a white cursor block.

```
Enter 25 numbers:
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25
Average = 13.00
...Program finished with exit code 0
Press ENTER to exit console.
```

5. Write a C program to find smallest array element in C.

PROGRAM :

```
#include <stdio.h>

int main() {

    int a[100], n, i, min;

    printf("Enter number of elements:\n");

    scanf("%d", &n);

    printf("Enter elements:\n");

    for(i = 0; i < n; i++) {

        scanf("%d", &a[i]);

    }

    min = a[0];

    for(i = 1; i < n; i++) {

        if(a[i] < min) {

            min = a[i];

        }

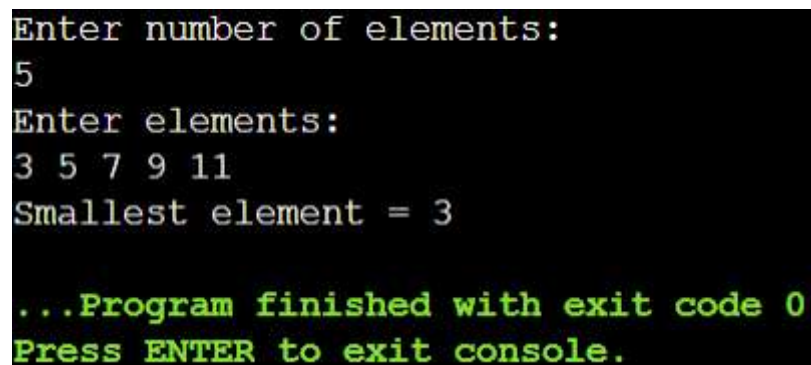
    }

    printf("Smallest element = %d", min);

    return 0;

}
```

OUTPUT:

A screenshot of a terminal window showing the execution of the C program. The text is as follows:

```
Enter number of elements:
5
Enter elements:
3 5 7 9 11
Smallest element = 3

...Program finished with exit code 0
Press ENTER to exit console.
```

6. Write a C Program to find Missing elements

PROGRAM :

```
#include <stdio.h>

int main() {

    int n, i, sum = 0, total;

    printf("Enter value of n:\n");

    scanf("%d", &n);

    int a[n-1];

    printf("Enter %d elements (one number missing):\n", n-1);

    for(i = 0; i < n-1; i++) {

        scanf("%d", &a[i]);

        sum += a[i];

    }

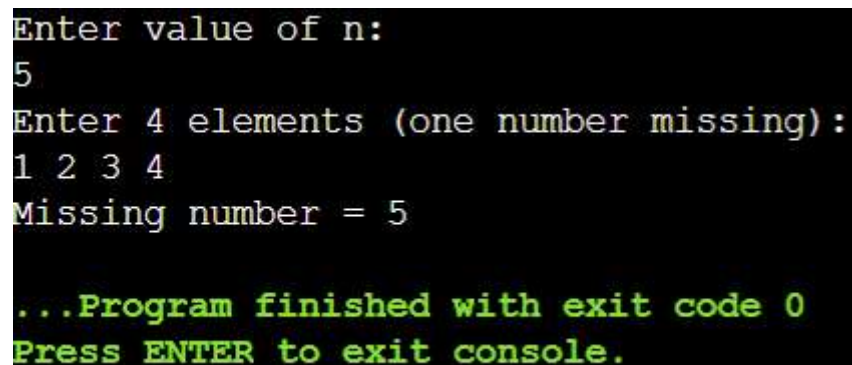
    total = n * (n + 1) / 2;

    printf("Missing number = %d", total - sum);

    return 0;

}
```

OUTPUT:

A screenshot of a terminal window showing the execution of the C program. The text is as follows:

```
Enter value of n:
5
Enter 4 elements (one number missing):
1 2 3 4
Missing number = 5

...Program finished with exit code 0
Press ENTER to exit console.
```

7. Write a C program to count the frequency of each element in an array.

PROGRAM :

```
#include <stdio.h>

int main() {

    int a[100], n, i, j, count, visited[100]={0};

    scanf("%d",&n);

    printf("Enter elements:\n");

    for(i=0;i<n;i++)

        scanf("%d",&a[i]);

    for(i=0;i<n;i++) {

        if(visited[i]) continue;

        count = 1;

        for(j=i+1;j<n;j++) {

            if(a[i]==a[j]) {

                count++;

                visited[j]=1;

            }

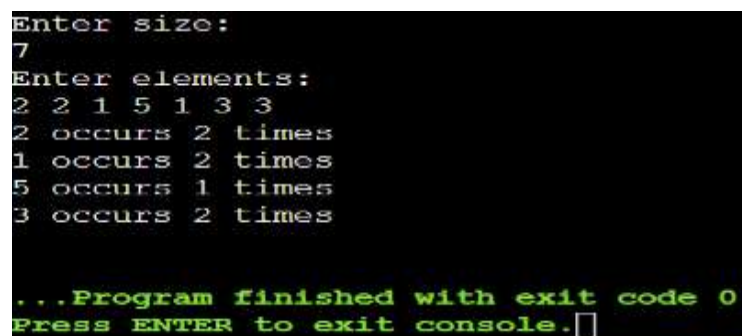
            printf("%d occurs %d times\n", a[i], count);

        }

        return 0;

    }
```

OUTPUT:



```
Enter size:
7
Enter elements:
2 2 1 5 1 3 3
2 occurs 2 times
1 occurs 2 times
5 occurs 1 times
3 occurs 2 times

...Program finished with exit code 0
Press ENTER to exit console.
```

8. Write a C program to find majority elements in an array..

PROGRAM :

```
#include <stdio.h>

int main() {

    int a[100], n, i, j, count;

    printf("Enter size:\n");

    scanf("%d",&n);

    printf("Enter elements:\n");

    for(i=0;i<n;i++)

        scanf("%d",&a[i]);

    for(i=0;i<n;i++) {

        count=0;

        for(j=0;j<n;j++)

            if(a[i]==a[j]) count++;

        if(count > n/2) {

            printf("Majority element = %d", a[i]);

            return 0;

        }

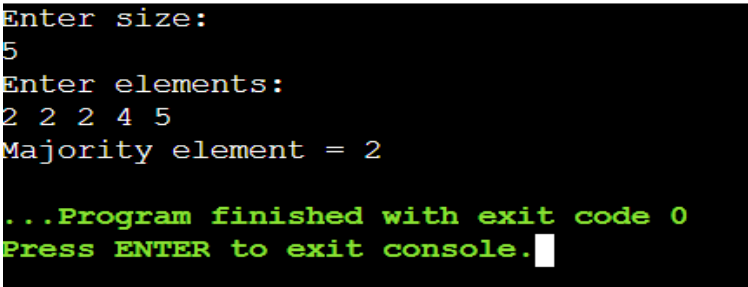
    }

    printf("No majority element");

    return 0;

}
```

OUTPUT:



```
Enter size:
5
Enter elements:
2 2 2 4 5
Majority element = 2

...Program finished with exit code 0
Press ENTER to exit console.
```


9. Write a C Program to delete an element in the array.

PROGRAM :

```
#include <stdio.h>

int main() {

    int a[100], n, i, pos;

    printf("Enter size:\n");

    scanf("%d",&n);

    printf("Enter elements:\n");

    for(i=0;i<n;i++)

        scanf("%d",&a[i]);

    printf("Enter position to delete:\n");

    scanf("%d",&pos);

    for(i=pos-1;i<n-1;i++)

        a[i]=a[i+1];

    printf("Updated array:\n");

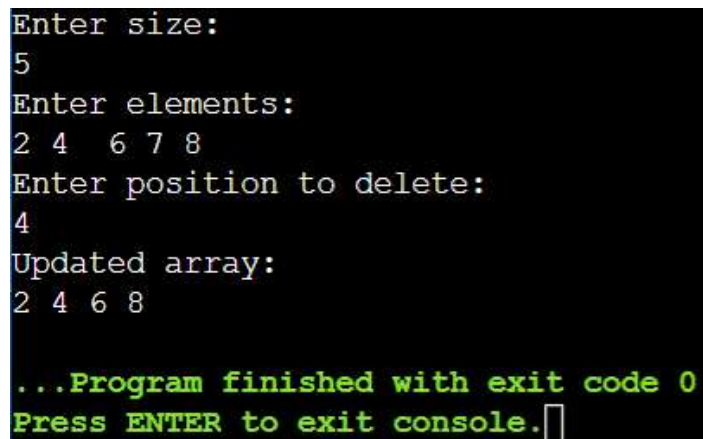
    for(i=0;i<n-1;i++)

        printf("%d ",a[i]);

    return 0;

}
```

OUTPUT:



```
Enter size:
5
Enter elements:
2 4 6 7 8
Enter position to delete:
4
Updated array:
2 4 6 8

...Program finished with exit code 0
Press ENTER to exit console.
```

10. Write a C Program to sort array in ascending order.

PROGRAM :

```
#include <stdio.h>

int main() {

    int a[100], n, i, j, temp;

    printf("Enter size:\n");

    scanf("%d",&n);

    printf("Enter elements:\n");

    for(i=0;i<n;i++)

        scanf("%d",&a[i])

    for(i=0;i<n;i++)

        for(j=i+1;j<n;j++)

            if(a[i]>a[j]) {

                temp=a[i];

                a[i]=a[j];

                a[j]=temp;

            }

    printf("Sorted array:\n");

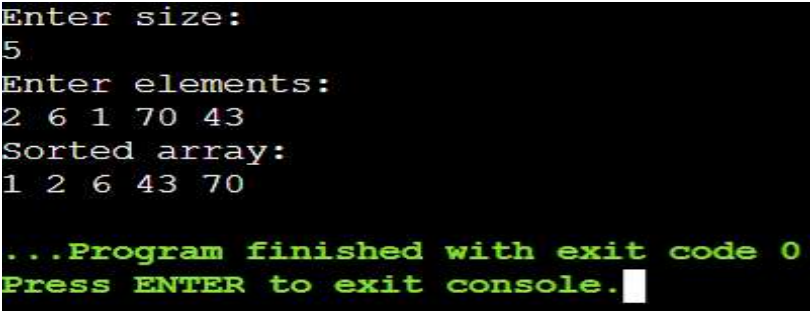
    for(i=0;i<n;i++)

        printf("%d ",a[i]);

    return 0;

}
```

OUTPUT:

A screenshot of a terminal window with a black background and white text. It shows the execution of a C program that sorts an array. The user enters '5' for the size and '2 6 1 70 43' for the elements. The program outputs the sorted array '1 2 6 43 70'. At the bottom, a green message states: '...Program finished with exit code 0 Press ENTER to exit console.' followed by a white cursor.

```
Enter size:
5
Enter elements:
2 6 1 70 43
Sorted array:
1 2 6 43 70

...Program finished with exit code 0
Press ENTER to exit console.
```

11. Write a program to find the common element(s) in two arrays.

PROGRAM :

```
#include <stdio.h>

int main() {

    int a[100], b[100];

    int n1, n2, i, j, found = 0;

    printf("Enter size of first array:\n");

    scanf("%d", &n1);

    printf("Enter elements of first array:\n");

    for(i = 0; i < n1; i++) {

        scanf("%d", &a[i]);

    }

    printf("Enter size of second array:\n");

    scanf("%d", &n2);

    printf("Enter elements of second array:\n");

    for(i = 0; i < n2; i++) {

        scanf("%d", &b[i]);

    }

    printf("Common elements are:\n");

    for(i = 0; i < n1; i++) {

        for(j = 0; j < n2; j++) {

            if(a[i] == b[j]) {

                printf("%d ", a[i]);

                found = 1;

                break;

            }

        }

    }

}
```

```
}  
if(found == 0) {  
    printf("No common elements");  
}  
return 0;  
}
```

OUTPUT:

```
Enter size of first array:  
3  
Enter elements:  
1 2 3  
Enter size of second array:  
5  
Enter elements:  
1 2 3 4 5  
Common elements:  
1 2 3  
  
...Program finished with exit code 0  
Press ENTER to exit console.
```

12. Write a C program to find all unique quadruplets in an array of sum up to a given value.

PROGRAM :

```
#include <stdio.h>

int main() {
    int a[50], n, i, j, k, l, sum;

    printf("Enter size:\n");

    scanf("%d", &n);

    printf("Enter elements:\n");

    for(i=0; i<n; i++) scanf("%d", &a[i]);

    printf("Enter required sum:\n");

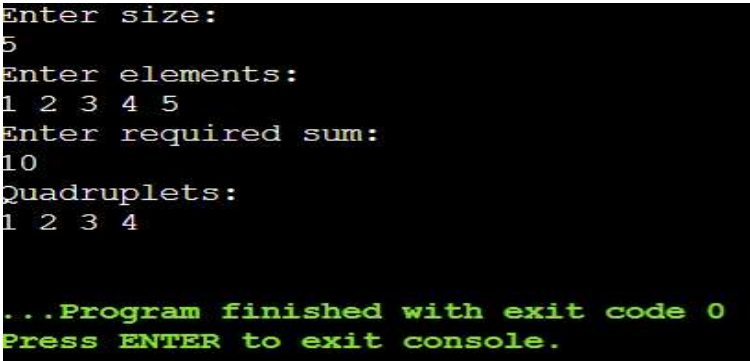
    scanf("%d", &sum);

    printf("Quadruplets:\n");

    for(i=0; i<n; i++)
        for(j=i+1; j<n; j++)
            for(k=j+1; k<n; k++)
                for(l=k+1; l<n; l++)
                    if(a[i]+a[j]+a[k]+a[l]==sum)
                        printf("%d %d %d %d\n", a[i], a[j], a[k], a[l]);

    return 0;
}
```

OUTPUT:



```
Enter size:
5
Enter elements:
1 2 3 4 5
Enter required sum:
10
Quadruplets:
1 2 3 4

...Program finished with exit code 0
Press ENTER to exit console.
```

13. Write a C program to find the triplets in an array whose sum equals a given value.

PROGRAM :

```
#include <stdio.h>

int main() {
    int a[50], n, i, j, k, sum;

    printf("Enter size:\n");

    scanf("%d", &n);

    printf("Enter elements:\n");

    for(i=0; i<n; i++) scanf("%d", &a[i])

    printf("Enter required sum:\n");

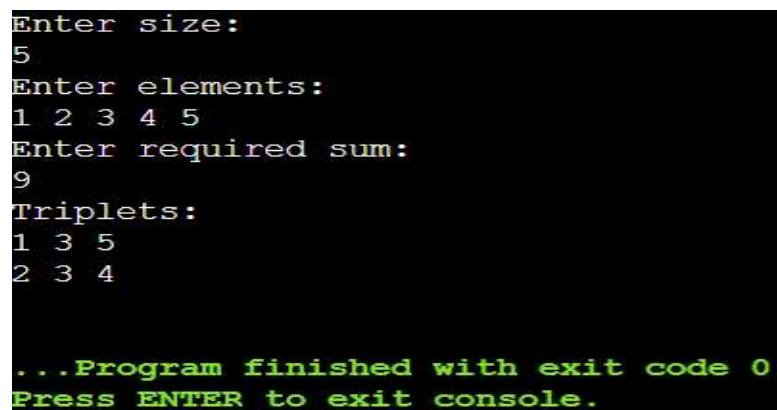
    scanf("%d", &sum)

    printf("Triplets:\n");

    for(i=0; i<n; i++)
        for(j=i+1; j<n; j++)
            for(k=j+1; k<n; k++)
                if(a[i]+a[j]+a[k]==sum)
                    printf("%d %d %d\n", a[i], a[j], a[k]);

    return 0;
}
```

OUTPUT:

A screenshot of a terminal window showing the execution of a C program. The program prompts the user to enter the size of the array, the elements, and the required sum. It then displays the triplets found in the array whose sum equals the required value. The output shows that for an array of size 5 with elements [1, 2, 3, 4, 5] and a required sum of 9, the triplets (1, 3, 5) and (2, 3, 4) are found. The program ends with a message indicating it finished with exit code 0 and prompts the user to press ENTER to exit the console.

```
Enter size:
5
Enter elements:
1 2 3 4 5
Enter required sum:
9
Triplets:
1 3 5
2 3 4

...Program finished with exit code 0
Press ENTER to exit console.
```

14. Write a C program to find the triplets in an array whose sum equals a given value.

PROGRAM :

```
#include <stdio.h>

int main() {
    int a[50], n, i, j, k, sum;

    printf("Enter size:\n");

    scanf("%d", &n);

    printf("Enter elements:\n");

    for(i=0; i<n; i++) scanf("%d", &a[i])

    printf("Enter required sum:\n");

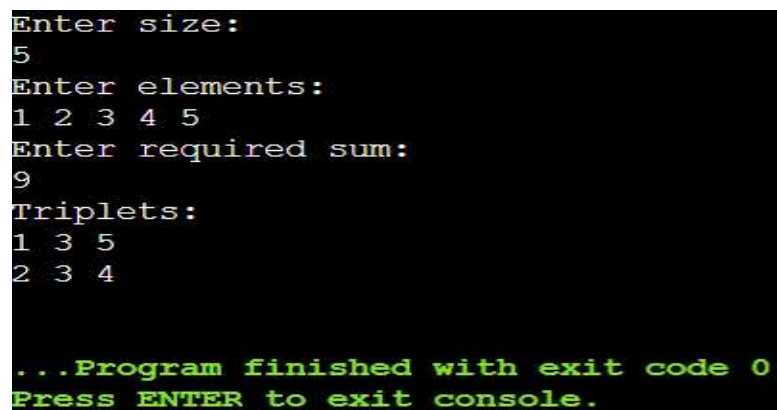
    scanf("%d", &sum)

    printf("Triplets:\n");

    for(i=0; i<n; i++)
        for(j=i+1; j<n; j++)
            for(k=j+1; k<n; k++)
                if(a[i]+a[j]+a[k]==sum)
                    printf("%d %d %d\n", a[i], a[j], a[k]);

    return 0;
}
```

OUTPUT:

A screenshot of a terminal window showing the execution of the C program. The user enters '5' for the size, '1 2 3 4 5' for the elements, and '9' for the required sum. The program outputs the triplets '1 3 5' and '2 3 4'. At the bottom, it shows the program finished with exit code 0 and prompts the user to press ENTER to exit the console.

```
Enter size:
5
Enter elements:
1 2 3 4 5
Enter required sum:
9
Triplets:
1 3 5
2 3 4

...Program finished with exit code 0
Press ENTER to exit console.
```

15. Write a C program to find the second largest element in an array.

PROGRAM :

```
#include <stdio.h>

int main() {

    int a[100], n, i, max, second;

    printf("Enter size:\n");

    scanf("%d",&n);

    printf("Enter elements:\n");

    for(i=0;i<n;i++) scanf("%d",&a[i]);

    max = second = -9999;

    for(i=0;i<n;i++) {

        if(a[i] > max) {

            second = max;

            max = a[i];

        } else if(a[i] > second && a[i] != max) {

            second = a[i];

        }

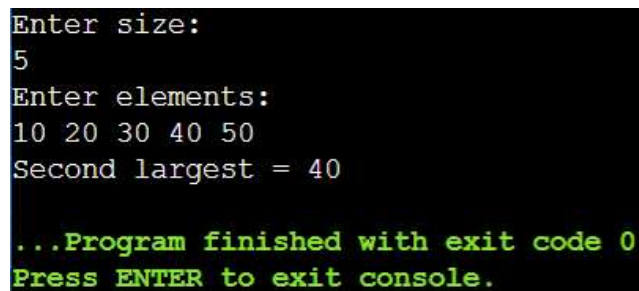
    }

    printf("Second largest = %d", second);

    return 0;

}
```

OUTPUT:

A screenshot of a terminal window showing the execution of the C program. The user enters '5' for the size and '10 20 30 40 50' for the elements. The program outputs 'Second largest = 40'. At the bottom, a green message states '...Program finished with exit code 0' and 'Press ENTER to exit console.'

```
Enter size:
5
Enter elements:
10 20 30 40 50
Second largest = 40

...Program finished with exit code 0
Press ENTER to exit console.
```


16. Write a C program to count all the pairs with a given difference.

PROGRAM :

```
#include <stdio.h>

int main() {

    int a[50], n, i, j, diff;

    printf("Enter size:\n");

    scanf("%d",&n);

    printf("Enter elements:\n");

    for(i=0;i<n;i++) scanf("%d",&a[i]);

    printf("Enter difference:\n");

    scanf("%d",&diff);

    printf("Pairs:\n");

    for(i=0;i<n;i++) {

        for(j=i+1;j<n;j++) {

            if((a[i]-a[j]==diff) || (a[j]-a[i]==diff))

                printf("(%d,%d)\n",a[i],a[j]);

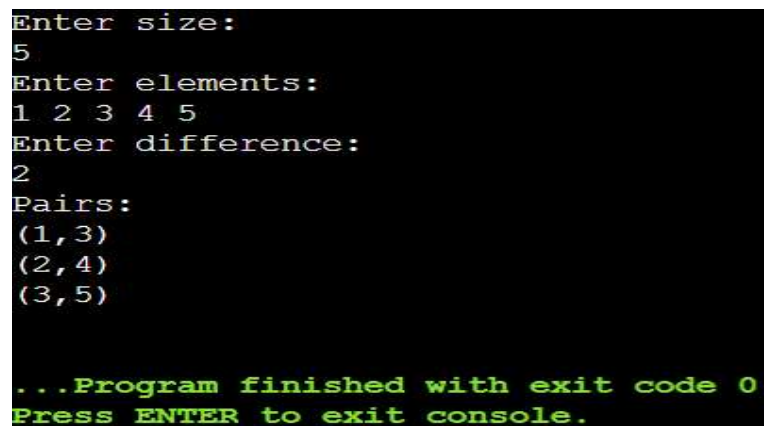
        }

    }

    return 0;

}
```

OUTPUT:

A screenshot of a terminal window showing the execution of the C program. The user enters '5' for size, '1 2 3 4 5' for elements, and '2' for difference. The program outputs three pairs: (1,3), (2,4), and (3,5). At the bottom, it says '...Program finished with exit code 0' and 'Press ENTER to exit console.'

```
Enter size:
5
Enter elements:
1 2 3 4 5
Enter difference:
2
Pairs:
(1,3)
(2,4)
(3,5)

...Program finished with exit code 0
Press ENTER to exit console.
```

17.write a program to find the union of two arrays.

PROGRAM :

```
#include <stdio.h>

int main() {

    int a[50], b[50], n, m, i, j, found;

    scanf("%d",&n);

    for(i=0;i<n;i++) scanf("%d",&a[i]);

    scanf("%d",&m);

    for(i=0;i<m;i++) scanf("%d",&b[i]);

    printf("Union:\n");

    for(i=0;i<n;i++)

        printf("%d ",a[i]);

    for(i=0;i<m;i++) {

        found = 0;

        for(j=0;j<n;j++) {

            if(b[i] == a[j]) {

                found = 1;

                break;

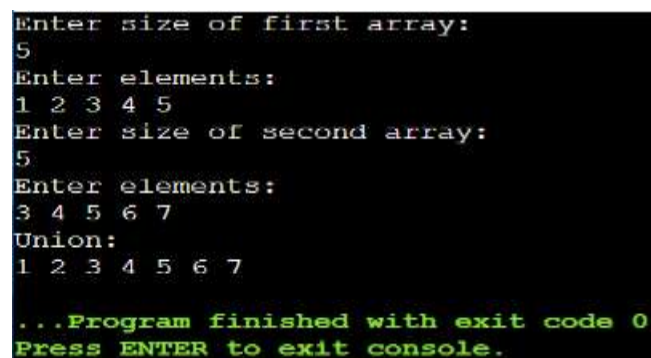
            } if(found == 0)

                printf("%d ",b[i]);

        } return 0;

    }
```

OUTPUT:



```
Enter size of first array:
5
Enter elements:
1 2 3 4 5
Enter size of second array:
5
Enter elements:
3 4 5 6 7
Union:
1 2 3 4 5 6 7

...Program finished with exit code 0
Press ENTER to exit console.
```

18. write a program to move all zeroes in an array to end while maintaining the order of elements.

PROGRAM :

```
#include <stdio.h>

int main() {
    int a[50], n, i, j = 0;

    printf("Enter size:\n");

    scanf("%d", &n);

    printf("Enter elements:\n");

    for(i = 0; i < n; i++) {
        scanf("%d", &a[i]);
    }

    for(i = 0; i < n; i++) {
        if(a[i] != 0) {
            a[j] = a[i];
            j++;
        }
    }

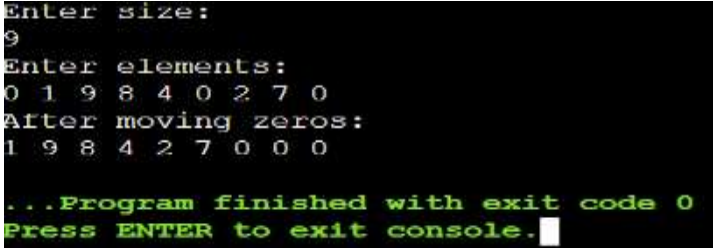
    while(j < n) {
        a[j] = 0;
        j++;
    }

    printf("Result:\n");

    for(i = 0; i < n; i++) {
        printf("%d ", a[i]);
    }

    return 0;
}
```

OUTPUT:



```
Enter size:
9
Enter elements:
0 1 9 8 4 0 2 7 0
After moving zeros:
1 9 8 4 2 7 0 0 0

...Program finished with exit code 0
Press ENTER to exit console.
```

19. Write a C program to count all the pairs with a given difference.

PROGRAM :

```
#include <stdio.h>

int main() {

    int a[50], n, i, j, diff, count = 0;

    printf("Enter size:\n");

    scanf("%d", &n)

    printf("Enter elements:\n");

    for(i = 0; i < n; i++) {

        scanf("%d", &a[i]);

    }

    printf("Enter difference:\n");

    scanf("%d", &diff);

    for(i = 0; i < n; i++) {

        for(j = i + 1; j < n; j++) {

            if(a[i] - a[j] == diff || a[j] - a[i] == diff) {

                count++;

            }

        }

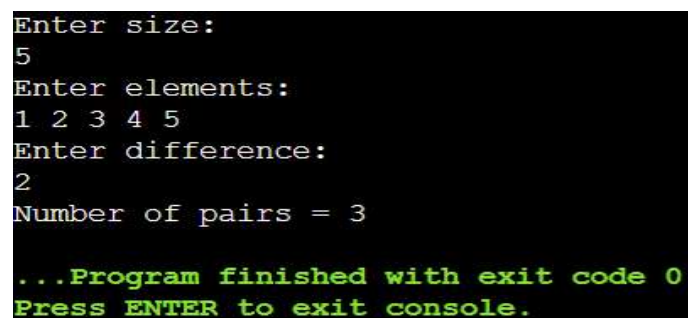
    }

    printf("Number of pairs = %d", count);

    return 0;

}
```

OUTPUT:

A screenshot of a terminal window showing the execution of the C program. The user enters '5' for size, '1 2 3 4 5' for elements, and '2' for difference. The program outputs 'Number of pairs = 3'. At the bottom, it says '...Program finished with exit code 0' and 'Press ENTER to exit console.' in green text.

```
Enter size:
5
Enter elements:
1 2 3 4 5
Enter difference:
2
Number of pairs = 3

...Program finished with exit code 0
Press ENTER to exit console.
```

20. Write a C Program to find the Kth smallest and Kth largest element in a given array.

PROGRAM :

```
#include <stdio.h>

int main() {

    int a[50], n, i, j, temp, k;

    scanf("%d", &n);

    printf("Enter elements:\n");

    for(i = 0; i < n; i++) {

        scanf("%d", &a[i]);

    }

    printf("Enter k:\n");

    scanf("%d", &k);

    for(i = 0; i < n; i++) {

        for(j = i + 1; j < n; j++) {

            if(a[i] > a[j]) {

                temp = a[i];

                a[i] = a[j];

                a[j] = temp;

            }

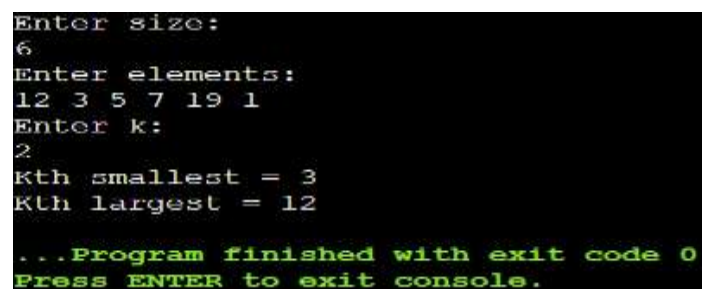
        }

    } printf("Kth smallest = %d\n", a[k-1]);

    printf("Kth largest = %d", a[n-k]);

    return 0;

} OUTPUT:
```

A screenshot of a terminal window showing the execution of the C program. The user enters '6' for the size, then '12 3 5 7 19 1' for the elements, and '2' for k. The program outputs 'Kth smallest = 3' and 'Kth largest = 12'. At the bottom, it says '...Program finished with exit code 0' and 'Press ENTER to exit console.'

```
Enter size:
6
Enter elements:
12 3 5 7 19 1
Enter k:
2
Kth smallest = 3
Kth largest = 12

...Program finished with exit code 0
Press ENTER to exit console.
```